

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2013-14

Zoology (Old Course)**Paper: ZOB-201 — Cell Biology and Biochemistry**

Time: Three hours

Max. Marks: 70

Note: Answer three questions from Section-A and three questions from Section-B. Question No. 1 in both sections is compulsory. Use a separate answer book for each section. Coloured sketches are allowed.

SECTION – A**(Cell Biology) (Marks: 35)****Question 1.** Answer the following parts:

(a) State and write the best option in the following:

[2 × $\frac{1}{2}$ = 1]

(i) Resolving power of a microscope is:

(1) $\frac{0.61\lambda}{\sin \theta}$

(2) $\frac{0.61\lambda}{\mu \sin \theta}$

(3) $\frac{\mu \sin \theta}{0.61}$

(4) $\frac{0.61}{\mu}$

(ii) Which one of the following is an intermediate filament?

(1) Protofilament

(2) Actin filament

(3) Myosin

(4) Keratins

(b) Write whether the following statements are True or False, giving reasons in 2–3 sentences each:

[3 × 2 = 6]

(i) Endoplasmic reticulum is involved only in lipid metabolism.

(ii) B cells are an essential component of the humoral immune response.

(iii) Polytene chromosome are found in amphibian oocytes.

(c) Define the following:

[2 × 1 = 2]

(i) Nucleosome

(ii) Transformed cell.

(d) Match the terms in Column-A with the best options in Column-B: $[4 \times \frac{1}{2} = 2]$

Column-A	Column-B
(i) Active transport	(1) Lysosome
(ii) Histone	(2) Na-K Pump
(iii) Rb	(3) Nucleosome
(iv) Multivesicular Bodies	(4) Tumor Suppression gene

Question 2. Answer the following parts:

- (a) What is cell theory? Explain the tenets of cell theory. $[4]$
- (b) Explain the fluid Mosaic Model of Plasma membrane with the help of labeled diagram. $[5]$
- (c) Illustrate the functions of Golgi body. $[3]$

Question 3. Answer the following parts:

- (a) What are cytoskeletons? Explain any cytoskeleton in detail. $[6]$
- (b) What is Chromatin? Explain how chromatin are organized in a nucleus. $[6]$

Question 4. Write notes on the following: $[3 \times 4 = 12]$

- (a) Functions of mitochondria
- (b) Biogenesis of ribosome
- (c) Lampbrush chromosome.

SECTION – B
(Biochemistry) (Marks: 35)

Question 1. Answer the following parts:

- (a) Indicate whether the following statements are true or false. Justify your answer in 2–3 sentences: [4 × 2 = 8]
- (i) Water molecules are held together by intra-ionic interactions.
 - (ii) Disulphide bonds are found in tertiary structure of proteins.
 - (iii) Amylopectin is a heteropolysaccharide.
 - (iv) The two strands of DNA are held together by phospho-diester bonds.
- (b) Draw structure of a lactose. [2]
- (c) Write Michaelis-Menten equation. [1]

Question 2. Answer the following parts:

- (a) With all possible details draw α -helix structure of a protein. [6]
- (b) Illustrate Hersey & Chase experiment to show that DNA serves as genetic material. [6]

Question 3. Answer the following parts:

- (a) Construct a Michaelis-Menten kinetic plot for an enzyme and define K_m of the enzyme. [6]
- (b) Illustrate stepwise events during transcription of a gene in prokaryotic system. [6]

Question 4. Write short notes on the following: [3 × 4 = 12]

- (a) Characteristics of a peptide bond
- (b) Genetic code
- (c) Phospholipids as component of lipid bi-layer.

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